

Brief Documentation (Tasks 1, 2 & Bonus Task 3)

This project demonstrates QA testing, API integration, and system monitoring using n8n automation workflows.

Task 1 – Web App QA Report

A demo web application was analyzed by testing core user flows such as sign-up, login, content creation, update, and deletion.

Work Done:

- Executed full application workflows
 - Identified **8 bugs** including:
 - API failure handling issues
 - Missing validation for undefined responses
 - CORS policy errors
 - Missing fallback UI states
 - Broken or missing assets
 - Registration and data loading failures
 - Documented each issue with:
 - Steps to reproduce
 - Expected vs actual results
 - Severity level
 - Root cause analysis (focus on missing error handling and validation)
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Task 2 – n8n API Integration Workflow

An automation workflow was built using n8n to integrate external APIs and send structured alerts.

APIs Used:

- **GitHub REST API** – to fetch repository data based on search query
- **Webhook.site** – to receive and verify alert outputs

Workflow Steps:

1. Schedule Trigger runs workflow at regular intervals
2. HTTP Request fetches data from GitHub API
3. Data transformation extracts key fields (repo name, stars, URL, timestamp)
4. IF condition filters valid results (e.g., minimum data or star threshold)
5. POST request sends structured JSON alert to webhook

Error Handling:

- Invalid or empty responses are filtered using IF conditions
 - Workflow does not crash on API failure
 - Only valid data is forwarded to alert system
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★ **Task 3 (Bonus) – Uptime Monitoring Workflow**

A second n8n workflow was created to monitor application health in real time.

Workflow Steps:

1. Schedule Trigger runs every 5 minutes
2. HTTP Request checks the web application status (Task 1 app)
3. IF condition checks HTTP status code
 - If status $\neq$ 200 $\rightarrow$ trigger alert
4. Alert is sent to the same notification channel used in Task 2

Bonus Enhancements:

- Retry logic added for failed requests
 - Response time tracking included in monitoring
 - Structured alert message with status, URL, and timestamp
 - Option for daily summary reporting (advanced enhancement)
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✂ **Summary**

This project demonstrates:

- QA testing and structured bug reporting (Task 1)
- API integration and workflow automation using n8n (Task 2)

- Real-time uptime monitoring with alerting system (Task 3 Bonus)

It combines testing, automation, and monitoring to simulate a real-world production observability pipeline.